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Filter Design

BOE071

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 7 | **Credits:** 3 | **L-T-P:** 3-0-0

Scheme: AKTU B.Tech 4th year, Open Elective-II, 2025-26 onwards

UNIT 1 Filter Fundamentals and Operational Amplifiers

UNIT 2 First-Order Filters

UNIT 3 Second-Order Low-Pass and Band-Pass Filters

UNIT 4 Second-Order Filters with Zeros and Butterworth Filters

UNIT 5 Chebyshev, Inverse Chebyshev and Cauer Filters

Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: transfer functions, circuits and comparison tables.
3. **Solved examples**: the numericals that appear in the paper (scaling, slew rate, gain-bandwidth, first-order RC design, Bode plots, Sallen-Key and biquad design, Butterworth and Chebyshev order and poles).
4. **Formulas / key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (filter types and specifications, why analog filters, circuit elements, scaling, op-amp models, slew rate, inverting and non-inverting circuits) and Unit 2 (bilinear transfer functions, first-order passive and active filters, Bode plots, finite gain effect, cascade design).
- **Afternoon (3 hours):** Unit 3 (second-order low-pass and band-pass: ω_o and Q, Sallen-Key, multiple feedback, integrators, Tow-Thomas and KHN biquads) and Unit 4 (transmission zeros: summing and feedforward; Butterworth response, poles, order).
- **Evening (2 hours):** Unit 5 (Chebyshev, inverse Chebyshev, elliptic/Cauer filters). Then the "Last-hour checklist".
- **In the exam:** always write the transfer function first, sketch the magnitude response with asymptotes, and show the specification diagram (passband, transition band, stopband with A_max and A_min).

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Filter Fundamentals and Operational Amplifiers

✍️ 2-mark definitions

- **Filter:** a circuit that passes signals in some frequency range (passband) and attenuates signals in another (stopband).
- **Transfer function H(s):** the ratio of output to input in the Laplace domain, V_o(s)/V_i(s), a ratio of polynomials N(s)/D(s).
- **Poles and zeros:** roots of the denominator and numerator of H(s). Poles must lie in the left half of the s-plane for stability.
- **Passband edge ω_p and stopband edge ω_s:** the frequencies bounding the passband and the stopband; the region between is the transition band.
- **A_max:** the maximum attenuation (ripple) allowed in the passband, in dB.
- **A_min:** the minimum attenuation required in the stopband, in dB.
- **Active filter:** a filter using op-amps (or other active devices) with resistors and capacitors, avoiding inductors and able to provide gain.
- **Passive filter:** a filter made only of R, L and C.
- **Frequency scaling:** changing the frequency of a normalized filter by a factor k_f, dividing all L and C values by k_f.
- **Impedance (magnitude) scaling:** multiplying all impedances by k_m: R → k_m R, L → k_m L, C → C/k_m; the transfer function is unchanged.
- **Slew rate (SR):** the maximum rate of change of an op-amp's output voltage, in V/µs.
- **Gain-bandwidth product (GBW, f_t):** for a single-pole op-amp, the product of closed-loop gain and bandwidth is constant and equal to the unity-gain frequency.

💡 Core concepts

Types of filters by response

Type	Passes	Example use
Low-pass (LP)	Below cutoff	Anti-aliasing before an ADC
High-pass (HP)	Above cutoff	Removing DC offset or hum
Band-pass (BP)	A band around ω_0	Radio channel selection
Band-stop / notch (BS)	All except a band	Removing 50 Hz mains hum
All-pass (AP)	All frequencies, changes phase	Delay and phase equalization

Types by approximation: Butterworth (maximally flat passband), Chebyshev (equal ripple in passband, sharper cutoff), inverse Chebyshev (flat passband, ripple in stopband), elliptic/Cauer (ripple in both, sharpest), Bessel (linear phase, constant delay).

Why analog filters are needed: signals in the real world are analog; anti-aliasing filters must come before an ADC and reconstruction filters after a DAC; very high frequencies (RF) are beyond digital processing; analog filters use less power and have no sampling delay in simple cases.

Circuit elements: resistors (thermal noise, tolerance), capacitors (prefer stable dielectrics such as C0G/NP0, polystyrene), inductors (bulky, lossy, not integrable at low frequency, hence active RC filters), op-amps.

Normalization and scaling: designs are tabulated for ω = 1 rad/s and 1 Ω; to get practical values, apply frequency scaling (ω_o new = k_f × 1) and impedance scaling (k_m): R_new = k_m R, C_new = C / (k_m k_f), L_new = k_m L / k_f.

Simulation and modelling: circuits are checked with SPICE (LTspice, Multisim) using AC sweeps for magnitude and phase, transient analysis for step response, Monte Carlo for tolerance, and op-amp macro-models.

Op-amp models

Model	Assumptions
Ideal	Infinite gain, infinite input impedance, zero output impedance, infinite bandwidth; virtual short between inputs
Finite DC gain	Gain A_o (about 10^5)
Single-pole (integrator) model	$A(s) = A_o\omega_a / (s + \omega_a) \approx \omega_t / s$, where $\omega_t = A_o\omega_a$ is the unity-gain frequency

Slew rate: limits large signals; full-power bandwidth f_max = SR / (2π V_p). Exceeding it distorts a sine into a triangle.

Resistive feedback circuits

Circuit	Gain
Inverting amplifier	$-R_f / R_1$
Non-inverting amplifier	$1 + R_f / R_1$
Voltage follower	1
Inverting summer	$-R_f (V_1/R_1 + V_2/R_2 + \ldots)$
Difference amplifier	$(R_2/R_1)(V_2 - V_1)$ with matched resistors

With a finite-bandwidth op-amp, the closed-loop bandwidth ≈ $\omega_t / (1 + R_f/R_1)$ for both inverting and non-inverting circuits (noise gain).

Applications of op-amps in filters: integrators (inverting, −1/(sRC)), differentiators, summers, buffers, gain stages, generalized impedance converters (simulated inductors).

🧮 Solved examples

Example 1: Scaling. A normalized low-pass prototype uses R = 1 Ω and C = 1 F (ω_o = 1 rad/s). Required f_o = 1 kHz with R = 10 kΩ.

- k_f = 2π × 1000 = 6283; k_m = 10^4.

- C_new = 1 / (10^4 × 6283) = **15.9 nF**. Check: f_o = 1 / (2π × 10^4 × 15.9 × 10^{-9}) = 1.0 kHz.

Example 2: Slew rate. An op-amp with SR = 0.5 V/µs must output a 10 V peak sine. f_max = 0.5 × 10^6 / (2π × 10) = **7.96 kHz**. Above that the output distorts.

Example 3: Gain-bandwidth. An op-amp with f_t = 1 MHz used as a non-inverting amplifier with gain 100 has bandwidth 1 MHz / 100 = **10 kHz**.

Key points to remember

- Filter types: LP, HP, BP, BS, AP. Approximations: Butterworth, Chebyshev, inverse Chebyshev, elliptic, Bessel.
- Specifications: ω_p, ω_s, A_max, A_min.
- Scaling: R × k_m, C ÷ (k_m k_f), L × k_m ÷ k_f.
- Op-amp: ideal, finite gain, single-pole ω_t/s; SR and full-power bandwidth; GBW.

⭐ Important questions

1. What is a filter? Classify filters and explain filter specifications. Why are analog filters needed?
2. Explain circuit elements and scaling (frequency and impedance) with an example.
3. Explain op-amp models, slew rate and gain-bandwidth product.
4. Explain resistive feedback op-amp circuits and their applications in filters.

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UNIT 2 First-Order Filters

✍️ 2-mark definitions

- **Bilinear transfer function:** a first-order transfer function $H(s) = (a_1 s + a_o) / (s + \omega_o)$, ratio of two first-order polynomials.
- **First-order low-pass:** $H(s) = K\,\omega_o / (s + \omega_o)$; slope −20 dB/decade above ω_o.
- **First-order high-pass:** $H(s) = K\,s / (s + \omega_o)$; slope +20 dB/decade below ω_o.
- **First-order all-pass:** $H(s) = (s - \omega_o)/(s + \omega_o)$ (or its negative): magnitude 1, phase changes from 0 to −180°.
- **Corner (cutoff, −3 dB) frequency:** the frequency where the magnitude is $1/\sqrt{2}$ of its passband value.
- **Bode plot:** a plot of magnitude in dB and phase against log frequency, drawn with straight-line asymptotes.
- **Cascade design:** realizing a high-order filter by connecting first- and second-order sections in series, each buffered by an op-amp so they do not load each other.

💡 Core concepts

Passive realizations: RC low-pass (series R, shunt C): $H = 1/(1 + sRC)$, $\omega_o = 1/RC$. RC high-pass (series C, shunt R): $H = sRC/(1 + sRC)$. RL versions are similar. Passive filters have gain ≤ 1 and depend on the load.

Active realizations

Circuit	Transfer function	Notes
Inverting low-pass (R_1 in, $R_2 \parallel C$ feedback)	$-(R_2/R_1) / (1 + sR_2C)$	Gain $-R_2/R_1$, $\omega_o = 1/R_2C$
Inverting high-pass (R_1 + C in series at input, R_2 feedback)	$-(R_2/R_1)\,s / (s + 1/R_1C)$	$\omega_o = 1/R_1C$
Non-inverting low-pass (RC then follower/amplifier)	$(1 + R_f/R_1) / (1 + sRC)$	Buffered output
All-pass (op-amp with RC on non-inverting input, equal R feedback)	$(1 - sRC)/(1 + sRC)$	Phase shifter

Bode plot rules: a pole at ω_o gives −20 dB/decade after ω_o and −45° at ω_o (phase from 0° to −90° over two decades); a zero gives the opposite. Actual magnitude at the corner is 3 dB below the asymptote.

Effect of finite op-amp gain: with a single-pole op-amp $A(s) = \omega_t/s$, an inverting low-pass picks up an extra pole near $\omega_t/(1 + R_2/R_1)$; the response departs from ideal near that frequency, the DC gain is slightly reduced (by the factor $1/(1 + (1 + R_2/R_1)/A_o)$). Keep ω_o much smaller than the closed-loop bandwidth.

Cascade design: H(s) = $H_1(s)$ $H_2(s)$ ... H_k(s); odd-order filters use one first-order section plus second-order sections. Order sections with increasing Q, keep gain distribution to avoid clipping and noise, and use op-amp outputs (low impedance) to drive the next section.

🧮 Solved examples

Example 1: First-order active low-pass. Design an inverting low-pass with DC gain −10 and f_o = 1 kHz using C = 10 nF.

- R_2 = 1 / (2π × 1000 × 10 × 10^{-9}) = **15.9 kΩ**.
- R_1 = R_2 / 10 = **1.59 kΩ**.
- H(s) = −10 × 6283 / (s + 6283).

Example 2: Bode plot values. For H(s) = 100 / (s + 100): DC gain = 1 (0 dB); at ω = 100 rad/s, |H| = 1/√2 (−3 dB), phase −45°; at ω = 1000, |H| ≈ 0.0995 (−20 dB), phase −84.3°; at ω = 10, phase −5.7°.

Example 3: All-pass phase. H = (1 − sRC)/(1 + sRC) has phase φ = −2 $\tan^{-1}$(ωRC). With R = 10 kΩ, C = 15.9 nF (RC = 159 μs), at 1 kHz ωRC = 1, so φ = **−90°**.

Example 4: Cascade of two identical first-order low-pass sections. Each has ω_o = 1000 rad/s. Overall H = $(1000/(s + 1000))^2$. At ω = 1000, each gives −3 dB, total −6 dB. The −3 dB frequency of the cascade is ω_o √(√2 − 1) = 0.644 × 1000 = **644 rad/s**, showing that cascading identical sections narrows the bandwidth.

Key points to remember

- Bilinear: $(a_1 s + a_o)/(s + \omega_o)$; LP, HP, AP special cases.
- RC low-pass ω_o = 1/RC; −20 dB/decade; −45° at ω_o.
- Active circuits: inverting LP gain $-R_2/R_1$ with $\omega_o = 1/R_2C$.
- Finite gain adds a parasitic pole near the closed-loop bandwidth.
- Cascade with buffered sections; n identical sections reduce the −3 dB frequency by √(2^(1/n) − 1).

⭐ Important questions

1. Explain the bilinear transfer function and its low-pass, high-pass and all-pass forms.
2. Explain passive and active realizations of first-order filters with circuits.
3. Draw the Bode plot of a first-order filter.
4. Explain the effect of finite op-amp gain on first-order filters.
5. Explain cascade design of filters.

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UNIT 3 Second-Order Low-Pass and Band-Pass Filters

✍️ 2-mark definitions

- **Biquadratic (biquad) transfer function:** $H(s) = (n_2s^2 + n_1s + n_0) / (s^2 + (\omega_o/Q)s + \omega_o^2)$.
- **Pole frequency ω_o:** the magnitude of the pole pair, the natural frequency.
- **Pole quality factor Q:** a measure of how close the poles are to the jω-axis; $Q = \omega_o / (2|\mathrm{Re}(p)|)$. High Q gives a sharper peak.
- **Bandwidth of a band-pass filter:** $BW = \omega_o / Q$ (between the −3 dB frequencies).
- **Second-order low-pass:** $H(s) = K\,\omega_o^2 / (s^2 + (\omega_o/Q)s + \omega_o^2)$.
- **Second-order band-pass:** $H(s) = K\,(\omega_o/Q)\,s / (s^2 + (\omega_o/Q)s + \omega_o^2)$.
- **Sallen-Key filter:** a single-op-amp second-order filter using positive feedback through a capacitor (VCVS topology).
- **Multiple feedback (MFB) filter:** a single-op-amp filter with two feedback paths to the inverting input.
- **Integrator:** an op-amp circuit whose output is the integral of the input: V_o = −(1/sRC) V_i.
- **Two-integrator-loop biquad:** a biquad built from two integrators in a loop (Tow-Thomas, KHN), giving independent tuning of ω_o and Q and several outputs.

💡 Core concepts

Design parameters: ω_o (centre or cutoff), Q (peaking or selectivity), K (gain). Pole positions $p = -\omega_o/(2Q) \pm j\omega_o\sqrt{(1 - 1/(4Q^2))}$. Q = 0.5: real repeated poles; Q = 0.707: Butterworth (maximally flat); Q greater than 0.707: peaking in the low-pass magnitude near ω_o, peak value ≈ Q for large Q.

Frequency response

Filter	At ω = 0	At ω = ω_o	At ω → ∞
Low-pass	K	K Q (magnitude)	Falls at −40 dB/decade
Band-pass	0	K (peak)	0, slopes ±20 dB/decade
High-pass	0	K Q	K

Sallen-Key low-pass (unity gain): R_1, R_2 in series to the non-inverting input; C_1 from the junction to the output (feedback), C_2 from the input to ground.

- $\omega_o = 1 / \sqrt{(R_1R_2C_1C_2)}$.
- $Q = \sqrt{(R_1R_2C_1C_2)} / (C_2(R_1 + R_2))$.
- With $R_1 = R_2 = R$: $Q = \frac{1}{2}\sqrt{(C_1/C_2)}$.
- Low sensitivity for low Q; widely used.

Equal-component Sallen-Key with gain K: $R_1 = R_2 = R$, $C_1 = C_2 = C$: $\omega_o = 1/RC$, $Q = 1/(3 - K)$; K must be less than 3 for stability, and Q is very sensitive to K near 3.

MFB band-pass: inverting, good for moderate Q (up to about 10–20); centre gain $-R_3/(2R_1)$ for equal capacitors, $Q = \frac{1}{2}\sqrt{(R_3/(R_1 \| R_2))}$.

Integrators: inverting (Miller) integrator $-1/(sRC)$; lossy integrator (R across C) gives a first-order low-pass; non-inverting integrator; phase-lead and lag errors from finite ω_t cause Q enhancement in biquads.

Two-integrator-loop biquads

- **KHN (state variable):** summer + two integrators; gives simultaneous high-pass, band-pass and low-pass outputs; $\omega_o = 1/RC$; Q set by a resistor ratio.
- **Tow-Thomas:** a lossy integrator, an ideal integrator and an inverter in a loop; low-pass and band-pass outputs; ω_o and Q set independently (Q by one resistor).
- Advantages: low sensitivity, easy tuning, high Q possible, multiple outputs; uses 3–4 op-amps.

🧮 Solved examples

Example 1: Sallen-Key Butterworth low-pass. $f_o = 1$ kHz, $Q = 0.707$, $R_1 = R_2 = 10$ kΩ, unity gain.

- $Q = \frac{1}{2}\sqrt{(C_1/C_2)} = 0.707 \rightarrow C_1/C_2 = 2$.
- $\sqrt{(C_1C_2)} = 1 / (2\pi \times 1000 \times 10^4) = 15.92$ nF.
- $C_2 = 15.92/\sqrt{2} =$ **11.25 nF**, $C_1 = 15.92 \times \sqrt{2} =$ **22.5 nF**.

Example 2: Pole positions. $\omega_o = 1000$ rad/s, $Q = 2$: $\mathrm{Re}(p) = -1000/4 = -250$; $\mathrm{Im}(p) = 1000\sqrt{(1 - 1/16)} = 968$. Poles at **−250 ± j968**. Low-pass peak ≈ Q × K = 2 (6 dB) near ω_o.

Example 3: Band-pass. A band-pass filter has $f_o = 10$ kHz and BW = 500 Hz. $Q = 10{,}000/500 =$ **20**. Lower and upper −3 dB frequencies ≈ $10{,}000 \mp 250 = 9.75$ kHz and 10.25 kHz (exactly $f_o(\sqrt{(1 + 1/4Q^2)} \mp 1/2Q)$).

Example 4: Equal-component Sallen-Key. For $Q = 0.707$: $K = 3 - 1/Q = 3 - 1.414 =$ **1.586** (non-inverting gain $1 + R_f/R_1$ with $R_f/R_1 = 0.586$).

Key points to remember

- Biquad denominator $s^2 + (\omega_o/Q)s + \omega_o^2$.
- BP bandwidth = ω_o/Q; LP magnitude at $\omega_o = KQ$.
- Sallen-Key: $\omega_o = 1/\sqrt{(R_1R_2C_1C_2)}$; equal R: $Q = \frac{1}{2}\sqrt{(C_1/C_2)}$; equal component: $Q = 1/(3 - K)$.
- KHN gives HP, BP, LP; Tow-Thomas gives LP and BP with independent ω_o, Q.

⭐ Important questions

1. Explain the design parameters ω_o and Q of second-order filters and their pole locations.
2. Explain second-order low-pass and band-pass circuits (Sallen-Key and MFB) and their frequency response.
3. Design a Sallen-Key low-pass filter for given specifications.
4. Explain integrators and two-integrator-loop biquads (KHN and Tow-Thomas).

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UNIT 4 Second-Order Filters with Zeros and Butterworth Filters

✍️ 2-mark definitions

- **Transmission zero:** a frequency where the filter output is zero; finite jω-axis zeros give notches ($H(j\omega_z) = 0$).
- **Notch filter:** a band-stop filter with zeros on the jω-axis, $H(s) = K(s^2 + \omega_z^2)/(s^2 + (\omega_o/Q)s + \omega_o^2)$.
- **Low-pass notch / high-pass notch:** notch with $\omega_z > \omega_o$ (LPN) or $\omega_z < \omega_o$ (HPN), used in elliptic filters.
- **Summing method:** realizing zeros by adding (with weights) the HP, BP and LP outputs of a biquad in a summer.
- **Voltage feedforward:** realizing zeros by feeding a fraction of the input forward into the integrators of a biquad, saving an op-amp.
- **Butterworth (maximally flat) response:** $|H(j\omega)|^2 = 1 / (1 + \varepsilon^2(\omega/\omega_p)^{(2n)})$; the first $2n - 1$ derivatives of the magnitude are zero at $\omega = 0$.

💡 Core concepts

Summing method: with KHN outputs V_HP, V_BP, V_LP, a summer forms $V_o = a\,V_{HP} + b\,V_{BP} + c\,V_{LP}$, giving numerator $a s^2 + b (\omega_o) s + c \omega_o^2$ (scaled) and so any biquad (notch with $b = 0$ and a, c chosen for $\omega_z^2 = c \omega_o^2/a$; all-pass with b negative).

Voltage feedforward (Tow-Thomas): inject the input through capacitors and resistors into each integrator and the inverter; the numerator coefficients are set by these feedforward elements; fewer op-amps than summing.

Butterworth low-pass

- Magnitude $|H(j\omega)| = 1 / \sqrt{(1 + (\omega/\omega_o)^{(2n)})}$ with ω_o the −3 dB frequency ($\varepsilon = 1$). For a general passband ripple A_{max}: $\varepsilon = \sqrt{(10^{(A_{max}/10)} - 1)}$.
- Roll-off −20n dB/decade; no ripple; monotonic.
- **Poles:** lie on a circle of radius ω_o (normalized 1) in the left half-plane, equally spaced by π/n: $p_k = \omega_o e^{(j(\pi/2 + (2k - 1)\pi/2n))}$, $k = 1 \ldots n$.
- Each complex pair gives a second-order section with $Q_k = 1 / (2 \sin((2k - 1)\pi/2n))$.
- **Order:** $n \geq \log[(10^{(A_{min}/10)} - 1)/(10^{(A_{max}/10)} - 1)] / (2 \log(\omega_s/\omega_p))$.
- Butterworth filters have no finite transmission zeros (all zeros at infinity); adding jω-axis zeros sharpens the cutoff but changes the type.

Normalized Butterworth polynomials ($\omega_o = 1$)

n	Polynomial
1	$s + 1$
2	$s^2 + 1.414s + 1$

n	Polynomial
3	$(s + 1)(s^2 + s + 1)$
4	$(s^2 + 0.765s + 1)(s^2 + 1.848s + 1)$
5	$(s + 1)(s^2 + 0.618s + 1)(s^2 + 1.618s + 1)$

🧮 Solved examples

Example 1: Butterworth order. Specs: A_max = 1 dB up to 1 kHz, A_min = 40 dB at 5 kHz.

- $(10^4 - 1)/(10^{0.1} - 1) = 9999 / 0.2589 = 38{,}622$.
- $n \geq \log(38{,}622) / (2 \log 5) = 4.587 / 1.398 = 3.28$, so **n = 4**.

Example 2: Butterworth third-order poles and Q. n = 3: poles at angles 120°, 180°, 240° on the unit circle: −1 and −0.5 ± j0.866. The quadratic section $s^2 + s + 1$ has Q = 1. Polynomial $(s + 1)(s^2 + s + 1) = s^3 + 2s^2 + 2s + 1$.

Example 3: Fourth-order Butterworth section Q values. $Q_1 = 1/(2 \sin 22.5°) = 1/0.765 =$ **1.307**; $Q_2 = 1/(2 \sin 67.5°) = 1/1.848 =$ **0.541**. Realize as two cascaded Sallen-Key sections (low Q first is common for dynamic range).

Example 4: Notch. To remove 50 Hz hum with a notch of Q = 5: $\omega_z = \omega_o = 2\pi \times 50 = 314$ rad/s, $H(s) = (s^2 + 314^2)/(s^2 + 62.8\, s + 314^2)$. Rejection band (−3 dB) width = 50/5 = 10 Hz.

Key points to remember

- Zeros by summing (extra op-amp) or feedforward (no extra op-amp).
- Notch numerator $s^2 + \omega_z^2$; LPN if $\omega_z > \omega_o$, HPN if $\omega_z < \omega_o$.
- Butterworth: maximally flat, poles on a circle, $Q_k = 1/(2 \sin((2k - 1)\pi/2n))$.
- Order formula with log ratio.

⭐ Important questions

1. Explain second-order filters with arbitrary transmission zeros: the summing method and voltage feedforward.
2. Explain notch filters (LPN, HPN).
3. Explain the Butterworth low-pass response, its pole locations and maximally flat property.
4. Find the order of a Butterworth filter from specifications and design it with cascaded sections.

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UNIT 5 Chebyshev, Inverse Chebyshev and Cauer Filters

✍️ 2-mark definitions

- **Chebyshev polynomial C_n(x):** $C_n(x) = \cos(n \cos^{-1} x)$ for $|x| \le 1$ and $\cosh(n \cosh^{-1} x)$ for $|x|$ greater than 1; $C_0 = 1$, $C_1 = x$, $C_{n+1} = 2x\,C_n - C_{n-1}$.
- **Chebyshev (type I) filter:** equal ripple in the passband, monotonic stopband: $|H(j\omega)|^2 = 1 / (1 + \varepsilon^2 C_n^2(\omega/\omega_p))$.
- **Ripple factor ε:** $\varepsilon = \sqrt{(10^{A_{max}/10} - 1)}$.
- **Inverse Chebyshev (type II) filter:** flat (monotonic) passband and equal ripple in the stopband, with finite jω-axis zeros.
- **Cauer (elliptic) filter:** equal ripple in both passband and stopband; the sharpest transition for a given order; uses Chebyshev rational (elliptic) functions.
- **Chebyshev rational function:** the rational function $R_n(\omega)$ used in elliptic filters, which equioscillates in both bands.

💡 Core concepts

Chebyshev magnitude: oscillates between 1 and $1/\sqrt{(1 + \varepsilon^2)}$ in the passband (n/2 ripples), then falls as roughly $20 \log(\varepsilon\, 2^{n-1}\, \omega^n)$ dB in the stopband. At $\omega = 0$ the magnitude is 1 for odd n and $1/\sqrt{(1 + \varepsilon^2)}$ for even n.

Pole locations (normalized $\omega_p = 1$): poles lie on an ellipse. With $a = (1/n) \sinh^{-1}(1/\varepsilon)$:

- $\sigma_k = -\sinh a \times \sin((2k - 1)\pi/2n)$, $\omega_k = \cosh a \times \cos((2k - 1)\pi/2n)$, $k = 1 \ldots n$.

Order: $n \ge \cosh^{-1}(\sqrt{((10^{A_{min}/10} - 1)/(10^{A_{max}/10} - 1))}) / \cosh^{-1}(\omega_s/\omega_p)$.

Butterworth vs Chebyshev

Feature	Butterworth	Chebyshev
Passband	Maximally flat	Equal ripple
Transition	Gradual	Sharper for same order
Order for given specs	Higher	Lower
Pole locus	Circle	Ellipse
Pole Q	Lower	Higher (more sensitive)
Phase / delay	Better	Worse (more overshoot)

Inverse Chebyshev: $|H|^2 = \varepsilon^2 C_n^2(\omega_s/\omega) / (1 + \varepsilon^2 C_n^2(\omega_s/\omega))$. Poles are reciprocals of Chebyshev poles; zeros on the jω-axis at $\omega_s / \cos((2k - 1)\pi/2n)$. Same order as Chebyshev for the same specs, but flat passband and better delay; realized with LPN sections.

Cauer (elliptic) filters: specified by n, A_{max}, A_{min} and ω_s/ω_p; poles and zeros come from tables or software (MATLAB `ellip`). Lowest order for given specs; poles and zeros realized as LPN/HPN biquads in cascade. Disadvantages: worst phase linearity, higher pole Q, more sensitivity.

Design procedure (any approximation): (1) normalize specs, (2) find order n, (3) find poles (and zeros) from formulas or tables, (4) group into second-order sections (pair each pole pair with the nearest zero pair for elliptic), (5) choose circuits (Sallen-Key, MFB, Tow-Thomas, KHN), (6) frequency and impedance scale, (7) simulate and tune.

🧮 Solved examples

Example 1: Chebyshev order. Same specs as Unit 4 (A_max = 1 dB at 1 kHz, A_min = 40 dB at 5 kHz).

- √38,622 = 196.5; $\cosh^{-1}(196.5) = 5.97$; $\cosh^{-1}(5) = 2.29$.
- n ≥ 5.97/2.29 = 2.61, so **n = 3** (Butterworth needed 4).

Example 2: Third-order Chebyshev poles (1 dB ripple). ε = √(10^0.1 − 1) = 0.5088; a = (1/3) $\sinh^{-1}(1/0.5088) = 0.476$; sinh a = 0.494, cosh a = 1.115.

- k = 1: σ = −0.494 × sin 30° = −0.247, ω = 1.115 × cos 30° = 0.966.
- k = 2: σ = −0.494, ω = 0.
- Poles: **−0.247 ± j0.966 and −0.494** (normalized to $\omega_p = 1$).
- Quadratic section: $\omega_o = \sqrt{0.247^2 + 0.966^2} = 0.997$, Q = 0.997 / (2 × 0.247) = **2.02**. First-order section at 0.494.

Example 3: Stopband check. At $\omega = 5\omega_p$, a third-order Chebyshev with ε = 0.5088 gives attenuation $10 \log(1 + \varepsilon^2 \cosh^2(3 \cosh^{-1}5))$ = **47.8 dB**, meeting 40 dB. A third-order Butterworth (−3 dB at ω_p) gives only $10 \log(1 + 5^6) = 41.9$ dB at $5\omega_p$, and with a 1 dB passband edge it falls short, which is why n = 4 was needed.

Example 4: Chebyshev polynomial. $C_2(x) = 2x^2 - 1$; $C_3(x) = 4x^3 - 3x$. $C_3(5) = 500 - 15 = 485$ (= $\cosh(3 \cosh^{-1} 5)$).

Formulas

Item	Formula
Ripple factor	ε = √(10^(A_max/10) − 1)
Butterworth order	n ≥ log[(10^(A_min/10) − 1)/(10^(A_max/10) − 1)] / (2 log(ω_s/ω_p))
Chebyshev order	n ≥ $\cosh^{-1}$√(...) / $\cosh^{-1}(\omega_s/\omega_p)$
Chebyshev poles	$\sigma_k = -\sinh a \sin \theta_k$, $\omega_k = \cosh a \cos \theta_k$, $\theta_k = (2k - 1)\pi/2n$, $a = \sinh^{-1}(1/\varepsilon)/n$
Butterworth Q	$Q_k = 1/(2 \sin \theta_k)$
Biquad Q from poles	$Q = \sqrt{\sigma^2 + \omega^2}/(2\lvert\sigma\rvert)$

⭐ Important questions

1. Explain Chebyshev polynomials and the equal-ripple magnitude response.
2. Find the pole locations of a Chebyshev filter and design it. Compare Butterworth and Chebyshev filters.

3. Explain inverse Chebyshev filters: specifications, pole and zero locations and design.
4. Explain Cauer (elliptic) filters and Chebyshev rational functions.

Last-hour checklist

☐ Filter types and approximations; specs ω_p, ω_s, A_max, A_min; why analog filters.

☐ Scaling (15.9 nF example); op-amp models; slew rate (7.96 kHz); GBW (10 kHz); resistive feedback circuits.

☐ Bilinear functions; first-order active LP design (15.9 kΩ, 1.59 kΩ); Bode values; all-pass phase; cascade (644 rad/s).

☐ Biquad form; ω_o and Q; pole positions; Sallen-Key design (11.25 nF, 22.5 nF); equal-component K = 1.586; MFB; KHN and Tow-Thomas.

☐ Zeros by summing and feedforward; notch (50 Hz).

☐ Butterworth magnitude, poles, polynomials table, order (n = 4), section Q (1.307, 0.541).

☐ Chebyshev polynomials, order (n = 3), poles (−0.247 ± j0.966, −0.494), Q 2.02; comparison table.

☐ Inverse Chebyshev and Cauer filters; design procedure.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Design of Analog Filters (2nd Indian ed.)	Rolf Schaumann, Haiqiao Xiao, Mac E. Van Valkenburg	Oxford University Press	WorldCat
2	Applications and Design with Analog Integrated Circuits (2nd ed.)	J. Michael Jacob	Pearson	WorldCat
3	Continuous-Time Active Filter Design	T. Deliyannis, Yichuang Sun, J. K. Fidler	CRC Press	WorldCat

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